DAY 12 | HANDS-ON ACTIVITIES

GenAI Intro, LLMs and Tokens: Problem Statements

Seven problems. Every notebook cell runs offline; add an OPENAI_API_KEY to see real model behaviour. Offline results show how the code works, not real model quality.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Count the Tokens NEW (v4)

Level: Warm-up | **Concepts:** tokens, context window

Tokenise three texts of different lengths (a tweet, an email, a page of a policy) and estimate how many pages would fit into a 128,000 token context window.

Sample run / expected result

```
Tweet: 31 tokens
Email: 212 tokens
Policy page: 640 tokens
Approx. pages in 128k: 200
```

Hints

- Use tiktoken if available, otherwise about 4 characters per token.

Problem 2: Temperature Explorer

Level: Easy | **Concepts:** softmax, sampling

Given logits for 5 candidate next words, print the probabilities at temperature 0.2, 1.0 and 2.0, and sample 10 words at each temperature.

Sample run / expected result

```
T=0.2: 'sunny' 0.97
T=2.0: 'sunny' 0.38, 'rainy' 0.24 ...
```

Hints

- probs = exp(logits / T) / sum(exp(logits / T))

Problem 3: Attention by Hand

Level: Medium | **Concepts:** Q, K, V, softmax

Using small made-up vectors for the words in "bank of the river", compute the attention weights and show which word "bank" attends to most.

Sample run / expected result

```
bank attends most to: river (0.41)
```

Hints

- Make the vectors for bank and river similar on purpose.

Problem 4: Pick the Model NEW (v4)

Level: Easy | **Concepts:** reasoning vs standard, LLM vs SLM

For eight SmartDesk tasks (for example: classify a ticket, plan a network migration, summarise a call, draft a refund policy), choose light, default, top-tier or SLM, and give a one-line reason.

Sample run / expected result

```
classify ticket      -> SLM or light model: narrow, high volume
plan migration       -> default, high reasoning effort: multi-step
summarise call       -> light model: simple, cheap
```

Hints

- Ask: how hard, how often, how private, how fast?

Problem 5: Token Bill Calculator NEW (v4)

Level: Medium | **Concepts:** input, output, reasoning, cached tokens

Write cost(input_t, output_t, cached_t, reasoning_t) using the placeholder prices in the notebook. Compare the cost per 1,000 requests for a short answer, a long answer and a reasoning answer.

Sample run / expected result

```
short     (1200 in, 150 out)          : 0.00180 per request
long      (1200 in, 1200 out)         : 0.00600 per request
reasoning (1200 in, 300 out, 2000 rsn) : 0.01040 per request
```

Hints

- Reasoning tokens are billed as output.
- Label every price as a placeholder.

Problem 6: Your First API Call

Level: Medium | **Concepts:** OpenAI API, keys, usage

Call the model through llm() for three prompts and print the answer, input tokens and output tokens. Show that the cell still runs with no key (offline demo mode).

Sample run / expected result

```
Mode: OFFLINE demo
[offline demo] Summarise: my bill doubled ... | in=12 out=9
```

Hints

- Never print or hard-code the key.
- Read the model from LLM_MODEL.

Problem 7: SmartDesk Summariser API

Level: Stretch | **Concepts:** FastAPI, token log, cost

Wrap the summariser in a FastAPI POST /summarise endpoint. Log every call to a CSV with ticket id, input tokens, output tokens and estimated cost. Test it with TestClient.

Sample run / expected result

```
POST /summarise -> 200 {"summary": "...", "usage": {...}}
token_log.csv: 3 rows
```

Hints

- Use the Pydantic Ticket model from Day 5.
- Return 422 for an empty body.